Averaging Scheme for the Aerosol and Carbon Detection LiDAR Onboard DaQi-1 Satellite

Xifeng Cao, Xingying Zhang, Lu Zhang, Minqiang Zhou, Yuhan Jiang, Jiqiao Liu, Tiantao Cheng, Chuncan Fan, Cheng Chen, and Lin Chen

***Abstract*— Atmospheric carbon dioxide (CO_2) is the primary anthropogenic driver of climate change, accounting for more than half of the total effective radiative forcing (ERF). Active remote-sensing technique using differential absorption light detection and ranging (LiDAR) (DIAL) is recognized as the most promising remote sensing means for atmospheric CO_2 measurements. The aerosol and carbon detection LiDAR (ACDL) instrument onboard the DaQi-1 (DQ-1) is dedicated to quantifying the global spatial distribution of atmospheric CO_2. To meet the requirement of accuracy and precision, a reasonable averaging scheme for ACDL measurements is needed to minimize the effect of random noise of observations on CO_2 retrievals. In this study, three averaging schemes were conducted in the retrieval process: averaging of CO_2 columns (AVX), averaging of differential absorption optical depth (AVD), and averaging of signals (AVS). The performances were compared at three representative sites. The experiments were first carried out on simulations. The results show that the optimal size of the averaging window is 50 km, corresponding to an averaging of measurements over 150 pulse pairs. In addition, the AVX and AVD schemes are less affected by altitude variations and can be applied to surfaces with moderate and severe topographic variation, such as hills and mountains, whereas the AVS method is more suitable for surfaces with slight topographic variation, such as oceans, plains, and terraces. Furthermore, the ACDL observations were also retrieved by applying three averaging schemes and validated against ground-based Total Carbon Column Observing Network (TCCON) measurements at the Xianghe station. The AVS scheme exhibits better performance than the AVX and AVD methods with the lowest biases of less than 0.5 ppm, which is consistent with the simulation results.**

***Index Terms*— Aerosol and carbon detection light detection and ranging (LiDAR) (ACDL) instrument, averaging scheme, carbon dioxide (CO_2), random noise.**

I. Introduction

CARBON dioxide (CO_2), as one of the long-lived greenhouse gases (GHGs), is recognized as the most important GHG. Atmospheric CO_2 can affect the energy budget at the top of the Earth's atmosphere, responsible for approximately 63% of the total effective radiative forcing (ERF) [1], [2]. Anthropogenic CO_2 emissions have increased rapidly, from an average of 4.5 ± 0.7 GtC yr^{-1} in the 1960s to 10.9 ± 0.9 GtC yr^{-1} during 2010–2019 [3]. The main causes of CO_2 emissions are fossil fuel combustion and land-use change. Only approximately 46% of CO_2 emissions have accumulated in the atmosphere, increasing CO_2 concentrations by about 132 parts per million (ppm), while the rest has been taken up by natural carbon sinks in oceans and terrestrial biosphere [4]. However, significant uncertainties remain on the identity and mechanisms of the natural sinks and their response to climate change. Therefore, quantifying atmospheric CO_2 concentration with high precision and accuracy is essential to improve our understanding of the carbon cycle and enable better predictions of future CO_2 growth rates and climate change.

Ground and space-based instruments have been employed for CO_2 measurements for several years. These measurements provide a continuous record of CO_2 concentrations, allowing studies of long-term CO_2 trends on a local or global scale. Surface-based GHG networks, such as the Total Carbon Column Observing Network (TCCON) [5], [6], [7] and the Global Atmospheric Watch (GAW) network, can provide precise and accurate CO_2 observations, but they remain sparse and unevenly distributed in space and time. Furthermore, most of the sites are located far away from large point sources or strong sinks, hindering accurate estimates of CO_2 emissions. Spaceborne measurements offer

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Xifeng Cao is with the Department of Atmospheric and Oceanic Sciences and the Institute of Atmospheric Sciences, Fudan University, Shanghai 200438, China, also with the Chinese Academy of Meteorological Sciences, China Meteorological Administration, Beijing 100081, China, and also with the Key Laboratory of Radiometric Calibration and Validation for Environmental Satellites, National Satellite Meteorological Center (National Center for Space Weather), Innovation Center for FengYun Meteorological Satellite (FYSIC), China Meteorological Administration (CMA), Beijing 100081, China (e-mail: 21113020001@fudam.edu.cn).

Xingying Zhang, Lu Zhang, Yuhan Jiang, and Lin Chen are with the Key Laboratory of Radiometric Calibration and Validation for Environmental Satellites, National Satellite Meteorological Center (National Center for Space Weather), Innovation Center for FengYun Meteorological Satellite (FYSIC), China Meteorological Administration (CMA), Beijing 100081, China (e-mail: zxy@cma.gov.cn; zhanglu_nsmc@cma.gov.cn; jiangyuhan22@mails.ucas.ac.cn; chenlin@cma.gov.cn).

Minqiang Zhou is with the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing 100029, China (e-mail: minqiang.zhou@mail.iap.ac.cn).

Jiqiao Liu, Chuncan Fan, and Cheng Chen are with the Key Laboratory of Space Laser Communication and Detection Technology, Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Shanghai 201800, China (e-mail: liujiqiao@siom.ac.cn; fanchuncan@siom.ac.cn; chencheng@siom.ac.cn).

Tiantao Cheng is with the Department of Atmospheric and Oceanic Sciences and the Institute of Atmospheric Sciences, Fudan University, Shanghai 200438, China (e-mail: ttcheng@fudan.edu.cn).

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an opportunity to reveal the global distribution and variability of atmospheric CO_2. Both passive and active instruments have been developed to quantify atmospheric CO_2 concentrations. Passive remote sensing methods estimate CO_2 columns from observations of reflected sunlight in near-infrared CO_2 bands. The Greenhouse Gases Observing SATellite-2 (GOSAT-2) and Orbiting Carbon Observatory-2 (OCO-2) are designed using this approach [8]. OCO-2 measurements show higher precision than previous missions, with a single-sounding precision of around 0.5 ppm [9]. Subsequently, the Chinese Carbon Dioxide Observation Satellite Mission (TanSat), Fengyun-3D (FY-3D), and GaoFen-5 (GF-5) were also launched [10], [11], [12]. However, there are still some problems in CO_2 flux calculations. Data availability for passive spectrometers is very limited, less than 10%. Because the footprint is relatively large, the observations are easily contaminated by clouds or optically thick aerosols. Therefore, it has been difficult to quantify emissions from the tropics and high latitudes in the winter hemisphere, which exhibit substantial emissions during the year. Nocturnal measurements are unavailable for passive systems. In addition, atmospheric scattering varies with solar illumination and observation angles, contributing to spatially and temporally varying biases in CO_2 retrievals and, thus, flux uncertainty.

Active remote-sensing techniques using differential absorption light detection and ranging (LiDAR) (DIAL) will help to fill critical gaps in coverage. It can operate globally during both day and night, even in the presence of thin or scattered clouds, improving the measurement density and coverage. All measurements are obtained with a fixed observational geometry, so it has the potential to reduce measurement biases from aerosol scattering. Moreover, DIAL is particularly sensitive to the lower troposphere, where the CO_2 concentrations respond most strongly to anthropogenic emissions. European Space Agency (ESA) and National Aeronautics and Space Administration (NASA) proposed the LiDAR missions Advanced Space Carbon and Climate Observation of Planet Earth (A-SCOPE) and Active Sensing of CO_2 Emissions over Nights, Days and Seasons (ASCENDS), respectively [13], [14]. The missions aim to provide atmospheric CO_2 measurements with sufficient global coverage and high accuracy to constrain the fluxes to within 0.02 Pg C yr^{-1} on a scale of 1000 $\times$ 1000 km. Several integrated path differential absorption (IPDA) LiDARs for airborne CO_2 detection have been developed that operate in the 1.57- or 2.05-μm CO_2 absorption bands. A series of airborne campaigns were carried out to evaluate the capability of active CO_2 remote sensing measurements. NASA Goddard Space Flight Center (GSFC) designed a pulsed multiwavelength IPDA LiDAR as a candidate for the ASCENDS mission, and seven flight campaigns were performed in 2014 and 2016 over a variety of surface conditions. The results demonstrated that the weighted column-averaged mixing ratio of CO_2 (XCO_2) retrievals was consistent with the in situ measurements with a bias of less than 1 ppm [15]. Mao et al. [16] found that partial-column XCO_2 measurements can be made to cloud tops using an airborne pulsed multiwavelength IPDA LiDAR, which would help estimate emissions from surface carbon sources. The LaRC team tested the stability of the Intensity-Modulated Continuous-Wave (IM-CW) Laser Absorption Spectrometer (LAS) in a generally uniform CO_2 environment. The results showed that CO_2 column variations remained within 0.5 ppm during the flight over the Gulf of Mexico, and the accuracy and precision were less than 0.3 ppm at a 1-hour time scale [17]. The Shanghai Institute of Optics and Fine Mechanics developed a 1.57-μm airborne double-pulse IPDA LiDAR system, and three flight experiments were conducted. The horizontal gradients in XCO_2 from airborne observations were consistent with the in situ measurements, indicating the capability to estimate CO_2 fluxes [18]. The CO_2 retrievals were also compared with OCO-2 and the Carbon Tracker CO_2 datasets, showing a good agreement among them [19]. Zhang et al. [20] proposed a spectral energy model to improve the accuracy and robustness of XCO_2 retrieval. The standard deviations were significantly reduced compared with the conventional retrieval algorithm [20]. In addition, the aerosol and carbon detection LiDAR (ACDL) instrument, the first spaceborne IPDA LiDAR, onboard the DaQi-1 (DQ-1) satellite was successfully launched in April 2022. A total of one year and six months of data has been received until November 2023. The data are being tested to ensure the reliability of the measurements.

To harness the full potential of IPDA LiDAR in carbon flux monitoring, stringent requirements have been proposed. Research has indicated that a precision of 0.26%–1.3% and an accuracy of 1 ppm for CO_2 measurements are essential to reduce flux uncertainties substantially [21]. Therefore, a series of simulations of spaceborne IPDA measurements were performed to investigate potential sources of measurement errors. In general, the accuracy of measurements is related to the systematic errors arising from uncertainties in meteorological fields, air number density, and LiDAR range measurements. The precision is associated with the random errors mainly caused by speckle noise and detection noise of the instrument. Gong et al. [22] discussed the sensitivity of CO_2 measurement to atmospheric temperature and pressure uncertainties. Menzies and Tratt [23] focused on wavelength optimization for satelliteborne LiDAR to offer suitable optical depth and minimize the interference of water vapor (H_2O) and temperature on CO_2 detection while presenting a high-weighting function in the troposphere. Han et al. [24] studied the influence of the orbit altitude and solar radiation on the spaceborne IPDA LiDAR. Simulations revealed that a high-altitude orbit can lead to larger random errors and gaps between adjacent tracks [24]. Ehret et al. [21] found that medium-sized instruments in terms of telescope aperture (0.5–1.5 m) and laser average power (0.4–4 W) tended to have low random errors of 0.2% at 1.6 μm and 0.4% at 2.1 μm. Wang et al. [25] mapped the monthly mean global distribution of relative random errors (RREs) with 60% of the RRE within 0.25%. In conclusion, the random noise from the instrument has an impact on the precision of CO_2 measurements, and the effect of random noise can be minimized by instrument and orbit design. In addition, an averaging of measurements along the satellite flight track is common in the retrieval process to degrade the random

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errors. Singh et al. [26] evaluated the capacity of the space-based 2-μm IPDA LiDAR and discovered that measurement precision can reach 0.35 ppm and the accuracy was smaller than 0.3 ppm with a 10-s averaging time. However, there are several averaging schemes. Different averaging schemes during retrieval result in different CO_2 columns because the parameters being averaged are not identical. Thus, each averaging scheme has its characteristics and is suitable for different surfaces. Moreover, a systematic bias on CO_2 retrievals will be introduced by the averaging process due to the nonlinearity of the IPDA LiDAR measurement equation. Therefore, a study is needed to provide a reasonable averaging scheme for ACDL observations at different surfaces to reduce random errors for CO_2 measurements while minimizing systematic bias caused by the averaging process.

The main objective of this article is to compare the performance of the three typical different averaging schemes and provide a reasonable averaging method for ACDL observations. The optimal size of the averaging window is analyzed before the comparison of averaging schemes. Then, the comparison experiments are conducted both on simulations and ACDL observations at three representative sites. The system bias due to the nonlinearity is considered in the averaging process. The random and residual systematic errors are applied to indicate the capability of averaging methods at different surfaces. Note that this article concentrates on the random and systematic errors due to random noise, and the impact of other factors is not discussed here. An overview of the principle of IPDA measurements and averaging schemes is presented in Section II. The materials, experimental sites, and performance model are provided in Section III. In Section IV, the optimal size of the averaging window is discussed, and the comparison of three averaging schemes is analyzed. Finally, discussion and conclusions are given in Section V.

II. Overview of the Principles

A. IPDA Principle

The ACDL is a spaceborne IPDA LiDAR dedicated to the XCO_2 measurements. The IPDA LiDAR is similar to DIAL instruments. The IPDA LiDAR measures the transmission difference between two adjacent wavelengths referring to the online and offline wavelengths, respectively. The online wavelength is positioned at the offset from the peak of the CO_2 absorption line with significant absorption, whereas the offline is far away from the peak with less absorption, serving as a reference. Thus, the transmission is mostly sensitive to the CO_2 absorption. Moreover, the two wavelengths are carefully chosen, ensuring minimal interference with other trace gases and air molecules [27]. In contrast to a DIAL, the IPDA LiDAR relies on the backscattered signal from a hard target rather than atmospheric backscatter. As a result, the higher backscatter signal from these targets improves the measurement sensitivity and enables a system with a relatively smaller power-aperture product. The XCO_2 can be retrieved from the backscattered signal of the IPDA LiDAR in the nadir viewing direction [28].

According to the LiDAR equation, the backscattering signal of IPDA LiDAR can be expressed as

$$P\left(\lambda_{\mathrm{ON,OFF}}, r\right) = \frac{E\left(\lambda_{\mathrm{ON,OFF}}\right)}{\tau_{\mathrm{eff}}} \cdot \frac{A}{r^2} \cdot \frac{\rho}{\pi} \cdot T_{\mathrm{opt}} \cdot \ T_{\mathrm{atm}} \cdot \exp\left[-\tau_{\mathrm{CO_2}}\left(\lambda_{\mathrm{ON,OFF}}, r\right) - \tau_{\mathrm{atm}}\right] \tag{1}$$

where $P(\lambda_{\mathrm{ON,OFF}}, r)$ is the power of echo signals at the online and offline wavelengths, $E(\lambda_{\mathrm{ON,OFF}})$ represents the transmitted laser pulse energy, A represents the area of the telescope, r is the distance from the spaceborne platform to the hard target, and T_{opt} is the transmission of system optical efficiency. T_{atm} is the atmospheric transmission including scatter and absorption of clouds, aerosols, and atmospheric molecules except for CO_2. (ρ/π) is the surface reflectance in sr^{-1} defined as the reflected power per steradian toward the receiver divided by the incident power. τ_{eff} is the effective pulsewidth of the echo signals, accounting for the spreads of the reflected laser pulse in the time domain. $\tau_{\mathrm{CO_2}}$ $(\lambda_{\mathrm{ON,OFF}}, r)$ and τ_{atm} are the double-path integrated optical depth of CO_2 and other constituents in the atmosphere, respectively.

As the online and offline wavelengths are very close, the scattering and extinction properties of the other constituents can be regarded as identical. Thus, the differential optical depth difference (DAOD) of CO_2 between the online and offline can be expressed by the ratio of echo pulses $P(\lambda_{\mathrm{ON}}, r)$ and $P(\lambda_{\mathrm{OFF}}, r)$ normalized by the monitoring pulses $P_0(\lambda_{\mathrm{ON}})$ and $P_0(\lambda_{\mathrm{OFF}})$

$$\mathrm{DAOD} = -\frac{1}{2}\ln\frac{P(\lambda_{\mathrm{ON}}, r) \cdot P_0(\lambda_{\mathrm{OFF}})}{P(\lambda_{\mathrm{OFF}}, r) \cdot P_0(\lambda_{\mathrm{ON}})}. \tag{2}$$

The column-weighted dry air mixing XCO_2 can be derived from ACDL measurements using the following equation:

$$XCO_2 = \frac{\int_{p_{\mathrm{TOA}}}^{p_{\mathrm{surf}}} \rho_{\mathrm{CO_2}}(p, T)\mathrm{WF}(p, T)dp}{\int_{p_{\mathrm{TOA}}}^{p_{\mathrm{surf}}} \mathrm{WF}(p, T)dp} = \frac{\mathrm{DAOD}}{\mathrm{IWF}} \tag{3}$$

where p represents the pressure, T represents the temperature, $\rho_{\mathrm{CO_2}}(r)$ is the dry air volume mixing XCO_2, and p_{surf} and p_{TOA} denote the pressure of the hard target and top of the atmosphere, respectively. $\mathrm{WF}(p, T)$ is the weighting function depicting the relative contribution of an atmospheric layer at pressure p to XCO_2. IWF denotes the vertical integral of weighting functions along the optical path. For IPDA LiDAR, the weighting function demonstrates a higher value near the surface, indicating that IPDA LiDAR is more sensitive to CO_2 in the lower troposphere where sources and sinks cause the most prevalent perturbations to the background atmospheric CO_2. $\mathrm{WF}(p, T)$ is given by the following equation:

$$\mathrm{WF}(p, T) = \frac{\sigma_{\mathrm{on}}(p, T) - \sigma_{\mathrm{off}}(p, T)}{\left(1 + \frac{M_{\mathrm{H_2O}}}{M_{\mathrm{dryair}}}\rho_{\mathrm{H_2O}}(p, T)\right) \cdot \mathrm{g} \cdot \mathrm{M_{dryair}}}\mathrm{N_A} \tag{4}$$

where $\sigma_{\mathrm{on,off}}(p, T)$ represents the absorption cross section of online and offline, $\rho_{\mathrm{H_2O}}(r)$ is the dry-air volume mixing ratio of H_2O, g is the local acceleration due to gravity, $\mathrm{N_A}$ is the Avogadro's number, and M_{dryair} and $M_{\mathrm{H_2O}}$ are the average mass of a dry air molecule and H_2O molecule, respectively. The weighting function is mainly determined by the atmospheric parameters and $\sigma_{\mathrm{on,off}}(p, T)$.

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B. Averaging Schemes

The APD detector of IPDA LiDAR collects the incoming photon flux and converts it into measurable signal voltage as measurements. The actual voltage of the echo signal contains not only the effective echo pulse backscattered from the hard target but also random noise: detection noise, shot noise, and speckle noise. Thus, there will be a large uncertainty in the intensity measurement for a single echo pulse. The speckle noise is negligible owing to the large field of view [29]. Detection noise is the main source of random noise. For ACDL measurements, the detection noise is mainly thermal noise, which approximately follows a Gaussian distribution. In this article, the signal-to-noise ratio (SNR) of a single shot is estimated by the following equation:

$$\mathrm{SNR} = \frac{\mathrm{MRP}}{\sqrt{2\mathrm{e}M^2B(\mathrm{RFP}+i_d)+4k_BT_{\mathrm{dw}}B/R_L}} \tag{5}$$

where M represents the internal gain factor, R represents the detector's responsivity, P is the target return power, e is the elementary charge, B is the electrical bandwidth, F is the detector's excess noise factor, i_d denotes dark current noise density, k_B is Boltzmann's constant, T_{dw} is the absolute temperature, and R_L is the feedback resistor.

To meet the requirement of precision on CO_2 retrievals, an averaging of measurements along the satellite flight track is often required by spaceborne IPDA LiDAR, which enhances the echo signals. The equivalent SNR averaged over N_{shot} samples can be expressed as $\mathrm{SNR}_{\mathrm{int}} = \mathrm{SNR}(N_{\mathrm{shot}})^{1/2}$, which reduces the standard deviation on CO_2 retrievals by a factor of $(N_{\mathrm{shot}})^{1/2}$. For example, the single-shot SNRs of the online and offline wavelengths are about 23 and 56 when the target reflectance is 0.1 sr^{-1}, respectively. $\mathrm{SNR}_{\mathrm{int}}$'s are 276 and 672, respectively. The RRE of XCO_2 can be significantly reduced.

There are three typical averaging schemes, as shown in Fig. 1: averaging of CO_2 columns (AVX), averaging of differential absorption optical depth (AVD), and averaging of signals (AVS). The formulas are given by (6)–(8). Triangular brackets denote the arithmetic mean of parameters. Different horizontal averaging schemes result in different CO_2 retrievals. In addition, the averaging process will lead to a bias on CO_2 retrievals due to the nonlinearity of the IPDA LiDAR measurement equation. Fig. 2 demonstrates the distribution of echo signals and DAOD. The echo signals are assumed to be normally distributed, as shown in Fig. 2(a). However, the corresponding DAOD no longer follows the normal distribution due to the nonlinearity of the IPDA LiDAR measurement equation. The true DAOD is 0.659, whereas the mean DAOD is about 0.663. There is a deviation between them, which will result in a bias on CO_2 measurements. Fortunately, this part of the systematic error can be estimated and corrected during the retrieval process

$$\overline{XCO_2}^{\mathrm{AVX}} = <\frac{\mathrm{DAOD}}{\mathrm{IWF}}> \tag{6}$$

$$\overline{XCO_2}^{\mathrm{AVD}} = \frac{<\mathrm{DAOD}>}{<\mathrm{IWF}>} \tag{7}$$

$$\overline{XCO_2}^{\mathrm{AVS}} = \frac{-\frac{1}{2}\ln\frac{<P(\lambda_{\mathrm{ON}},r)><P_0(\lambda_{\mathrm{OFF}})>}{<P(\lambda_{\mathrm{OFF}},r)><P_0(\lambda_{\mathrm{ON}})>}}{<\mathrm{IWF}>}. \tag{8}$$

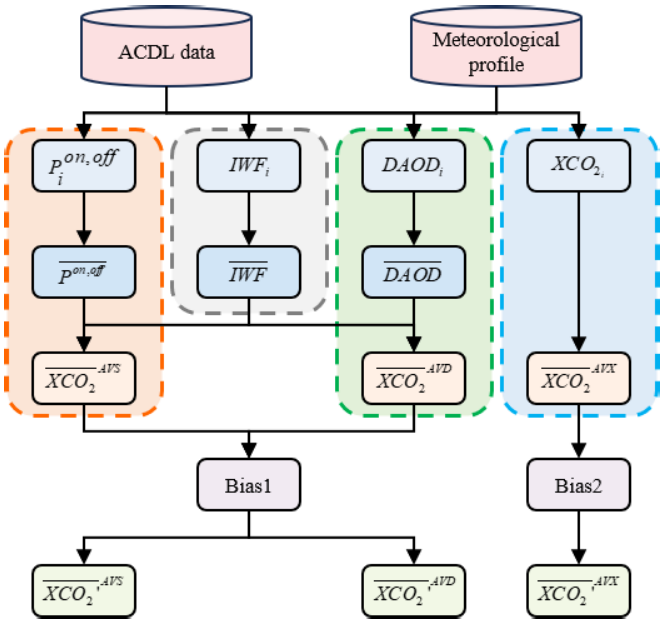


Fig. 1. Schematic for three averaging methods.

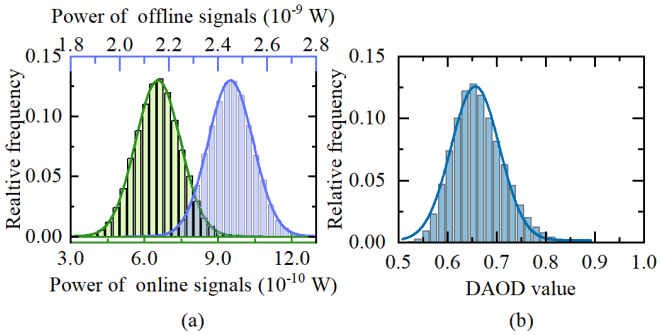


Fig. 2. Impact of the nonlinearity of the IPDA LiDAR measurement equation on the DAOD distribution for land surface with ice and snow cover. (a) Normally distributed echo signals at online and offline wavelengths. (b) Corresponding distribution of DAOD.

For the AVX method, each observation is affected by random noise, so there will be a bias in every DAOD measurement. The normalized echo signals of online and offline are assumed to be normally distributed around a mean value $\mu^{\mathrm{ON,\ OFF}}$ and with a standard deviation $\sigma^{\mathrm{ON,\ OFF}}$. Hence, the bias of AVX due to random noise can be reduced based on (9) and (10). For the AVD method, the bias is the same as that of the AVX method. Thus, the systematic error can also be estimated by the above two equations. More details about the principle of bias correction are described in [30]

$$\mathrm{SNR}^{\mathrm{on,\ off}} = \frac{\mu^{\mathrm{on,off}}}{\sigma^{\mathrm{on,off}}} \tag{9}$$

$$\mathrm{Bias} = \frac{1}{\sqrt{2\pi}}\int_{-\mathrm{SNR}^{\mathrm{OFF}}}^{+\infty}\ln\left(1+\frac{x}{\mathrm{SNR}^{\mathrm{OFF}}}\right)\cdot \mathrm{e}^{-\frac{x^2}{2}}dx - \frac{1}{\sqrt{2\pi}}\int_{-\mathrm{SNR}^{\mathrm{ON}}}^{+\infty}\ln\left(1+\frac{x}{\mathrm{SNR}^{\mathrm{ON}}}\right)\cdot \mathrm{e}^{-\frac{x^2}{2}}dx. \tag{10}$$

For the AVS method, the online and offline echo signals are first averaged separately, and then, the DAOD values are

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calculated. Therefore, the bias is for the averaged signals rather than a single shot, which is greatly decreased compared to AVX and AVD methods. The equivalent SNR of N-pulse averages can be given by (11), where N denotes the number of pulse pairs. $\mu_i^{\mathrm{ON,OFF}}$ and $\sigma_i^{\mathrm{ON,OFF}}$ represent the mean value and standard deviation of individual normalized signals, respectively,

$$\mathrm{SNR}_{\mathrm{int}}^{\mathrm{ON,OFF}} = \sqrt{N}\frac{\sum_{i=1}^{N}\mu_i^{\mathrm{ON,OFF}}}{\sqrt{\sum_{i=1}^{N}\left(\sigma_i^{\mathrm{ON,\ off}}\right)^2}}. \tag{11}$$

III. Materials and Methods

To compare the performance of three averaging schemes, the simulations are conducted. The comparisons are also carried out on the ACDL measurements. Satellite observations of Cloud-Aerosol LiDAR and Infrared Pathfinder Satellite Observations (CALIPSO), Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), Landsat 9, DQ-1, and European Centre for Medium-Range Weather Forecasts (ECMWF) analyses are used to constrain the simulation as close as possible to real-world conditions. The details of the dataset and performance model are discussed in the following.

A. Datasets

1) CALIPSO Dataset: The attenuation of clouds and aerosols is provided by the CALIPSO observations. CALIPSO is a dual-wavelength polarization LiDAR launched in April 2002 [31]. It is designed to study the impact of clouds and aerosols on climate change [32]. CALIPSO traveled in the A-train orbit at an altitude of 705 km and operated until September 2018. The measurements span from 82 °N to 82 °S. Thus, CALIPSO can provide global aerosol and cloud observation data during the day and night. The orbit repeat cycle is 16 days, and the equator-crossing time is about 1:30 A.M./P.M.

The total vertical optical depth by aerosol and cloud layers is calculated based on the Level2 product of Cloud-Aerosol LiDAR with Orthogonal Polarization (CALIOP) with a spatial resolution of 5 km. Level 2 product consists of a cloud optical depth of 532 nm and aerosol optical depths of 532 and 1064 nm. The total optical depth (TOD) is the sum of the optical depths of aerosols and clouds at 1064 and 532 nm, respectively. Although the wavelengths of 532 and 1064 nm are far away from 1572 nm, the influence of aerosols is almost negligible in the ACDL measurements.

2) Landsat-9 Dataset: The intensity of the IPDA echo signal is directly proportional to the surface reflectance, so the reflectance plays a significant role in the simulations. Surface reflectance is provided by Landsat 9 Level 2 product. Landsat 9 was launched successfully in November 2021 with a design life of eight years [33]. It is a Sun-synchronous satellite orbiting at an altitude of 705 km. The orbital inclination is about 98.2°. There are two payloads onboard the Landsat 9 satellite: Operational Land Imager 2 (OLI-2) and Thermal Infrared Sensor 2 (TIRS-2). OLI-2 consists of nine channels ranging from visible, near-infrared to short-wave infrared waves, which observe the global land surface at a moderate scale. The sixth band of the OLI-2 instrument is the short-wave infrared channel (1570–1650 nm), covering the 1572 nm applied for ACDL observations. Therefore, the Level 2 product of OLI-2 approximates the surface reflectance during the ACDL observations. The spatial resolution is about 30 m along a 185-km-wide swath. The OLI-2 L2 images were resampled to a spatial resolution of 90 m, which was comparable to the size of the footprint of ACDL measurements. The resampled latitude and longitude were regarded as the pseudomeasurement of the ACDL instrument.

3) DQ-1 Dataset: The Atmospheric Environmental Monitoring Satellite (AEMS), also known as the DQ-1 satellite, was successfully launched on April 16, 2022 [34]. Five payloads were onboard DQ-1, including Directional Polarization Camera-2 (DPC-2), Environment Monitoring Instrument-2 (EMI-2), ACDL, Particulate Observing Scanning Polarimeter (POSP), and Wide Spectral Imager (WSI). The DQ-1 satellite operates in a Sun-synchronous orbit with an altitude of 705 km. The orbit inclination is approximately 98.203°, allowing global coverage between 82 °N and 82 °S. The equator-crossing time is about 1:30 P.M. with a 51-day repeat cycle.

ACDL is the world's first spaceborne IPDA LiDAR instrument. The ACDL payload was designed by the Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Science, whereas the platform was developed by the China Aerospace Science and Technology Corporation (CNSA). The main objective of the ACDL instrument is to quantify the global distribution of atmospheric CO_2 with high accuracy. It operates in nadir mode with a divergence angle of 0.1 mrad, corresponding to a footprint of 70 m along the satellite track. The ACDL instrument consists of three channels: 1572 nm for CO_2 detection and 532 and 1064 nm for clouds and aerosols detection. The CO_2 columns are retrieved from the backscattered echo signals at 1572 nm. Dual pulses near 1572 nm are switched at 200-μs intervals to ensure the overlap of footprint between online (1572.024 nm) and offline (1572.085 nm) wavelengths. The pulse repletion rate is 20 Hz, which means that 20 measurements can be made in one second. The backscattered signals are collected by the receiver telescope with a diameter of 1000 mm and then recorded by the APD detector. Fig. 3 illustrates the optical design of the ACDL instrument.

4) Other Auxiliary Dataset: Surface elevations were derived from the Terra ASTER Global Digital Elevation Model (GDEM) version 3 (ASTGTM). The geographic coverage of the ASTER GDEM extends from 83 °N to 83 °S with a spatial resolution of approximately 30 m. The data are divided into 1° latitude by 1° longitude tiles and distributed in both Cloud Optimized GeoTIFF (COG) and NetCDF4 formats via NASA Earth data Search. The projection is the World Geodetic System 1984 (WGS84)/Earth Gravity Model 1996 (EGM96) geoid. The elevations of the pseudo-measurements were interpolated from the ASTGTM data.

CO_2 profiles were taken from the TCCON GGG2020 data and served as the true values of the CO_2 columns. The

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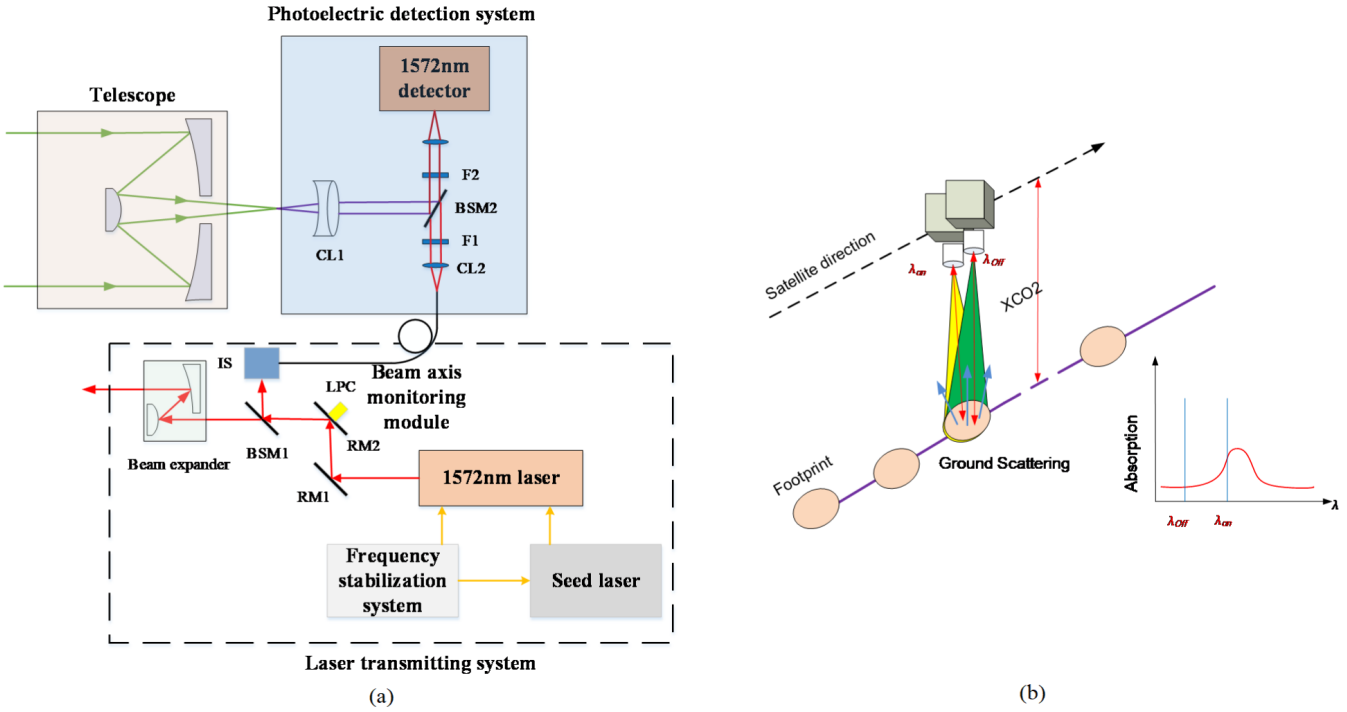


Fig. 3. Overview of the ACDL configuration. (a) Schematic of the ACDL optical system. (b) Observation mechanism of the ACDL instrument.

horizontal distribution of CO_2 was set to be identical for each layer during the simulations. Meteorological profiles such as temperature, pressure, and humidity profiles were derived from the ECMWF Reanalysis v5 (ERA5) dataset.

B. Experimental Sites

There are several averaging schemes, each of which is suitable for different situations. To analyze the applicability of the averaging methods, three sites with different topographical variability were selected. These sites were carefully selected in the same latitude area to reduce interference from the background atmosphere. The three cities are Fuyang, Huaihua, and Ya'an. The three scenes represent surface types with slight, moderate, and severe topographic variations, respectively. In addition, surface reflectance at the three sites was scaled according to the mean reflectance of the different land cover types to assess the impact of land cover types on the performance of averaging schemes. Four mean reflectance values were considered: 0.1 (vegetation), 0.05 (mixed sea and vegetation), 0.025 (sea and ocean), and 0.016 (ice and snow).

The characteristics of the three sites are summarized in Table I. The experimental sites encompass a wide range of altitude variations. The altitude standard deviation of Huaihua and Ya'an is approximately 226.8 and 567.4 m, respectively. Fig. 4(a) presents the elevation of each shot along the satellite track. Fuyang exhibits a relatively even distribution in elevation, ranging from 10 to 50 m, whereas Ya'an demonstrates a severe altitude variation, spanning from 160 to 950 m. Fig. 4(b) depicts the variation of relative reflectance at three experimental sites. The reflectance variation in the averaging window is what we are concerned with. Ya'an shows the most significant surface reflectance variation, followed by Huaihua, whereas Fuyang shows the lowest fluctuation.

TABLE I
CHARACTERISTICS OF THE THREE SITES

Parameters	Fuyang	Huaihua	Yaan
Latitude	33.08-33.53°N	29.72-30.17°N	30.44-30.88°N
Longitude	115.86°E	111.22°E	102.72°E
Altitude (m)	10.7-49.5	160.8-945.6	1120.1-3733.3
Mean altitude (m)	29.0	497.5	2528.1
Standard deviation of Altitude (m)	4.8	226.8	567.4
Surface pressure (hPa)	1007.3-1012.0	904.7-994.1	638.3-885.8
Mean surface pressure (hPa)	1009.8	955.2	745.8
Standard deviation of surface pressure (hPa)	0.6	25.9	53.9
Relative reflectance (sr^{-1})	0.40-1.36	0.29-1.81	0.39-2.47
Standard deviation of relative reflectance (sr^{-1})	0.12	0.33	0.43

C. Overall Framework

This article aims to test the applicability of different averaging methods at three sites with different topographic variations. Experiments were carried out on both simulations and ACDL observations. The detailed parameters of the spaceborne IPDA LiDAR system are shown in Table II. Simulations were conducted based on the above dataset and instrument parameters. Only the clear sky simulations are performed during the experiment. CO_2 columns may be obtained from the cloud-top reflections but were not

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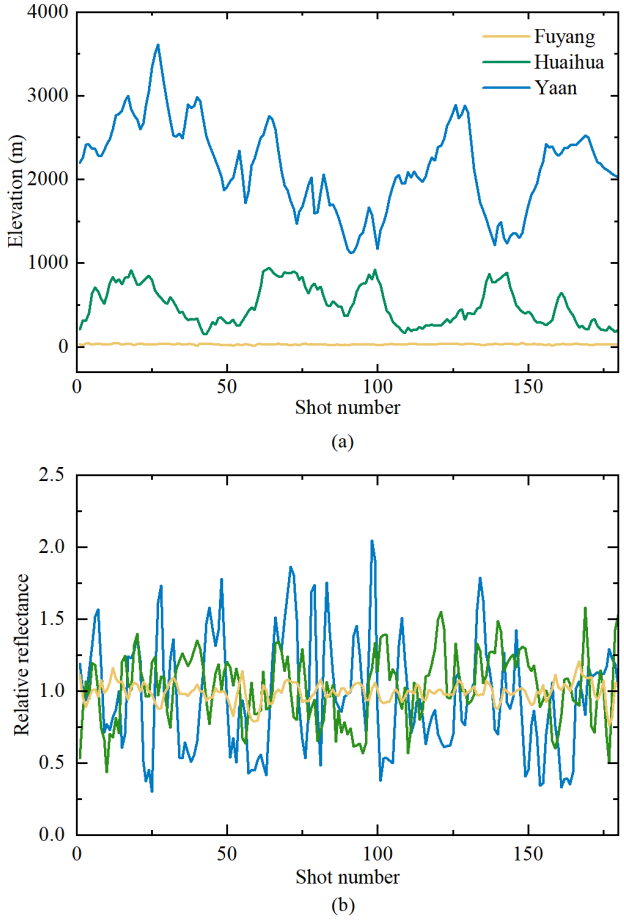


Fig. 4. (a) Elevation and (b) relative reflectances for each shot along the satellite track.

considered here. During the simulations, the absorption cross sections were evaluated based on line-by-line calculation and the High-Resolution Transmission Molecular Absorption (HITRAN) database 2020 version. A Voigt line shape function was assumed for each absorption line to account for collision and Doppler broadening effects during the simulation. Since the calculation of the cross section took a lot of time, a 4-D lookup table of the absorption cross section was created in advance to improve the efficiency of the simulations. Echo signals of online and offline wavelengths were simulated according to (1). Moreover, a random noise was added to the original signal. The SNR is calculated according to (5). In this analysis, the optical depths of H_2O were considered since it represented the major interfering atmospheric molecule. However, interferences from other atmospheric molecules, such as ozone (O_3), nitrous oxide (N_2O), and ammonia (NH_3), were neglected due to either weaker absorption around the 1.57-μm wavelength or lower abundance.

Based on the simulations, the optimal size of the averaging window was determined first, which involved a tradeoff between resolution and precision. Then, XCO_2 retrievals were carried out based on (3) and (4) with the different averaging schemes. The averaging bias due to the nonlinearity of the IPDA LiDAR measurement equation was also considered. The retrievals from different averaging schemes were validated

TABLE II
INSTRUMENT PARAMETERS FOR IPDA LIDAR

	Parameter	Value	Unit
Platform	Orbit altitude	700	Km
	Orbit type	Sun-synchronous	-
	Viewing geometry	Nadir	-
Transmitter	Online wavelength	1572.024	nm
	Offline wavelength	1572.085	nm
	Pulse energy	75	mJ
	Pulse width	15	ns
	Repetition rate	20	Hz
Receiver	Telescope diameter	1	m
	Field of view	0.2	mrad
	Optical efficiency	0.69	-
	Quantum efficiency	0.75	-
	Bandwidth	2	MHz
	Feedback resistance	1	MΩ
	Excess noise factor	5.5	-
	Noise equivalent power	35	$fw/\sqrt{Hz}$

against the true value, which is a weighted average column dry air volume mixing ratios converted from input CO_2 profiles using the same weighting function applied in the retrieval process. Finally, the performance of the three averaging methods was compared by evaluating the accuracy of the CO_2 columns. Furthermore, CO_2 retrievals were also conducted based on real ACDL measurements using different averaging methods to verify the accuracy of the bias correction and test the effectiveness of different averaging methods.

Since the detection noise is random, the deviation between CO_2 retrievals and true value must be assessed through a series of simulations to show the performance of the averaging methods. The number of simulations M must be high enough to evaluate the systematic error of CO_2 retrievals with sufficient accuracy. To get an estimate of the bias with an accuracy of 0.01 ppm with 95% confidence, approximately 80 000 samples are required according to the central limit theorem. Thus, 80 000 simulations were performed in this comparison experiment.

IV. RESULTS

A. Single Shot Measurements of CO_2 Columns

For IPDA LiDAR measurements, the accuracy and precision of XCO_2 retrievals are affected by random noise. Note that the accuracy and precision in this article refer to the systematic and random errors resulting from random noise, respectively. On the one hand, random noises cause a systematic bias in DAOD value due to the nonlinearity of the IPDA LiDAR measurement equation. On the other hand, random noises lead to fluctuations in the echo signals, resulting in random errors in XCO_2 retrievals. The accuracy and precision of the measurements are mainly determined by SNR, which, in turn, is affected by the surface reflectance and altitude of the hard target. To investigate the impact of surface reflectance and altitude on the accuracy and precision of CO_2 measurements, a sensitivity analysis of the CO_2 retrievals was performed based on simulations.

Fig. 5 presents the variation of systematic and random errors of a single shot. The systematic errors resulting from

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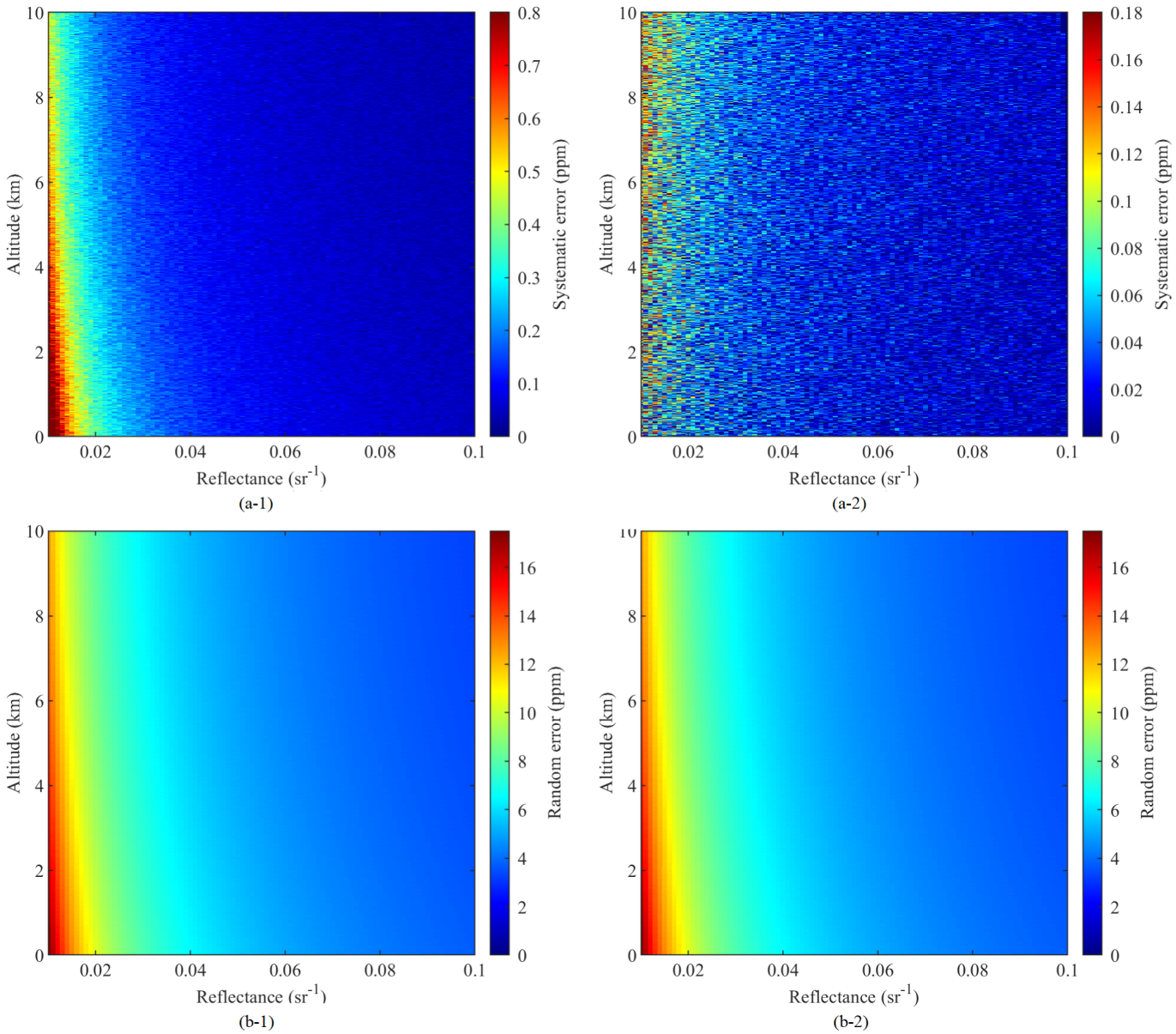


Fig. 5. Impact of surface reflectance and altitude on systematic and random errors of a single shot. (a-1) and (a-2) Systematic errors and residual errors after bias correction, respectively. (b-1) and (b-2) Same as (a-1) and (a-2) except for random errors.

the nonlinearity of the IPDA LiDAR measurement equation are inversely proportional to the surface reflectance and altitude, as shown in Fig. 5(a-1). This is because a higher reflectance results in a stronger LiDAR echo signal according to the LiDAR equation, while the hardware parameters remain unchanged and the detector operates in the linear region. Meanwhile, a stronger echo signal indicates a better SNR, enabling smaller random noise. Similarly, the strong echo signal is obtained from the hard target with a higher altitude due to less attenuation, consequently leading to a larger SNR. In addition, compared to the altitude of echo signals, the systematic errors are more sensitive to surface reflectance variations, which is most obvious when the altitude is lower. The single-shot systematic errors for ACDL simulations are basically in the range of 0–0.8 ppm. The systematic errors can be corrected, as explained in Section II. The residual systematic errors after bias correction are demonstrated in Fig. 5(a-2). The systematic errors decreased from about 0.35 to 0.08 ppm for surfaces with ice and snow cover. Although it is not completely corrected, the accuracy of XCO_2 retrievals is improved, which is meaningful for CO_2 measurements with high accuracy.

The variation of the random errors is demonstrated in Fig. 5(b-1). The random errors range from 1.5 to 19 ppm. The sensitivity of random errors follows a similar pattern to that of systematic errors. It is negatively correlated to the surface reflectance and the altitude of echo signals. The random errors after bias correction are the same as the original result. For the scattering surface of land cover, the random errors are approximately 3.3 ppm, which indicates that the precision of XCO_2 retrievals from a single shot does not meet the requirements of 1 ppm. Thus,

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an averaging of measurements along the satellite flight track is necessary.

B. Averaging Window Size

An averaging of measurements along the satellite flight track allows for a better SNR, which can improve the precision of retrievals. During the averaging process, the CO_2 retrievals depend not only on the averaging method but also on the number of averaged pulse pairs, also known as averaging window size. The effect of the averaging window size on the accuracy and precision of CO_2 measurements is analyzed, and the reasonable size of the averaging window is discussed.

The choice of averaging window size is guided by several criteria. First, it should effectively reduce random noise in the observations. Second, the errors introduced by variations in the CO_2 column should be of the same order of magnitude as other error sources. Finally, the averaging window size should enable ACDL measurements to yield a valid spatial distribution of CO_2 concentration for carbon source and sink studies. Simulations were conducted to analyze the impact of averaging window sizes on retrievals for various surface types to determine the optimal size. The variations in precision are similar for the three averaging methods. For simplicity, we only present the result of the AVD method here.

Fig. 6 demonstrates the distribution of the systematic and random errors with respect to the averaging window size. The systematic errors at Fuyang are depicted in Fig. 6(a-1) with different colors representing different surface reflectances. As the number of averaged pulse pairs increases, the systematic errors jitter a lot at the beginning and then gradually become stable. This is because the systematic errors of AVD methods are the average of the single-shot systematic errors. Thus, the more the number of pulses becomes, the more stable the systematic errors are. The systematic errors no longer decrease when the number of averaged pulse pairs exceeds 70, which corresponds to an average window size of 23.1 km. Furthermore, the averaging bias is inversely proportional to surface reflectance. The systematic errors are approximately 1.0, 0.5, 0.2, and 0.1 ppm for the surface with a reflectance of 0.1, 0.05, 0.025, and 0.016 sr^{-1}, respectively. Fig. 6(a-2) shows the residual errors after averaging bias correction. The accuracy has been significantly improved compared to Fig. 6(a-1). The residual errors for all surface reflectances are basically within 0.1 ppm. The lowest bias is found at a surface reflectance of 0.1 sr^{-1}, which was reduced to 0.01 ppm from 0.2 ppm after bias correction. The precision of the pulse pair averaged was presented in Fig. 6(a-3). Only the random errors after averaging bias correction are provided, as the difference between the precorrected and postcorrected precision is negligible. The random errors degrade with the number of averaged pulse pairs according to a power law $N^{-1/2}$. As the number of averaged pulse pairs approaches 150, the slope of the curve is about -0.01, indicating that the random errors no longer decrease significantly with increasing numbers. The precision for the pulse pair averaged is improved to better than 0.4 ppm on the surface with the reflectance of 0.1 sr^{-1}. In addition, precision is also related to surface reflectance. When the average window size is the same, the higher the surface reflectance is, the better the precision found. The surface reflectance also determines the upper limit of precision. For surfaces with the reflectance of 0.1, 0.05, 0.025, and 0.016 sr^{-1}, the precision is approximately 0.3, 0.5, 0.8, and 1.2 ppm, respectively.

The systematic and random errors for Huaihua exhibit a similar pattern to Fuyang, as shown in Fig. 6(b). The precision and accuracy become stable when the number of averaged pulse pairs exceeds 150. The residual systematic errors for all surface reflectance after bias correction are within 0.05 ppm. The precision of the 150-pulse pair averaged is obviously better than that of a single shot, within 0.45 ppm for the land surface. Fig. 6(c) shows the systematic and random errors for Ya'an. The systematic errors of Ya'an are slightly different from the two cities mentioned above. The systematic errors hardly converge either before or after bias correction. The residual systematic errors, after bias correction, initially decrease and then gradually increase in the negative direction. This is caused by the greater fluctuations in the CO_2 column due to the severe altitude variations. When the averaging window is small, the DAOD is similar and can be averaged to reduce the random errors of CO_2 retrievals. As the averaging window increases, the DAOD fluctuates dramatically with the severe altitude variations. Thus, a new geometric error is introduced into the CO_2 measurements for the Ya'an area. Fortunately, the percentage of scenes with severe altitude variations in the averaging window is significantly lower than with slight altitude variations. Only in a few areas were the CO_2 column affected by strong altitude variations.

In general, an averaging of measurements over 150 pulse pairs in the CO_2 retrieval proves to be more appropriate, corresponding to an averaging window size of about 50 km. The precision is significantly improved to within 0.36% for all surface types, and the spatial resolution meets the requirements of carbon cycle studies.

C. Comparison of Averaging Schemes for Simulations

Three typical averaging schemes were conducted in the retrieval process: AVX, AVD, and AVS. The performance of three averaging schemes was compared based on the same simulations. The accuracy of the CO_2 columns was applied to indicate the applicability.

Fig. 7(a) depicts the systematic errors of each averaging scheme under three surface types. It should be noted that the systematic bias due to random noise is still included in the results, as it is not yet subtracted. The mean reflectance is taken as a typical value of 0.05 sr^{-1}. Different colors represent different averaging methods. The AVX and AVD schemes exhibit the same systematic errors in all three scenarios, indicating a similar performance. The systematic errors of the AVX, AVD, and AVS schemes are approximately 0.18, 0.18, and -0.01 ppm in Fuyang, respectively. The systematic errors of the AVX and AVD schemes are relatively higher than the AVS scheme in Fuyang, where the altitude variations are low. The low systematic errors of AVS are mainly attributed to the high equivalent SNR of the averaged signals. AVS averaging

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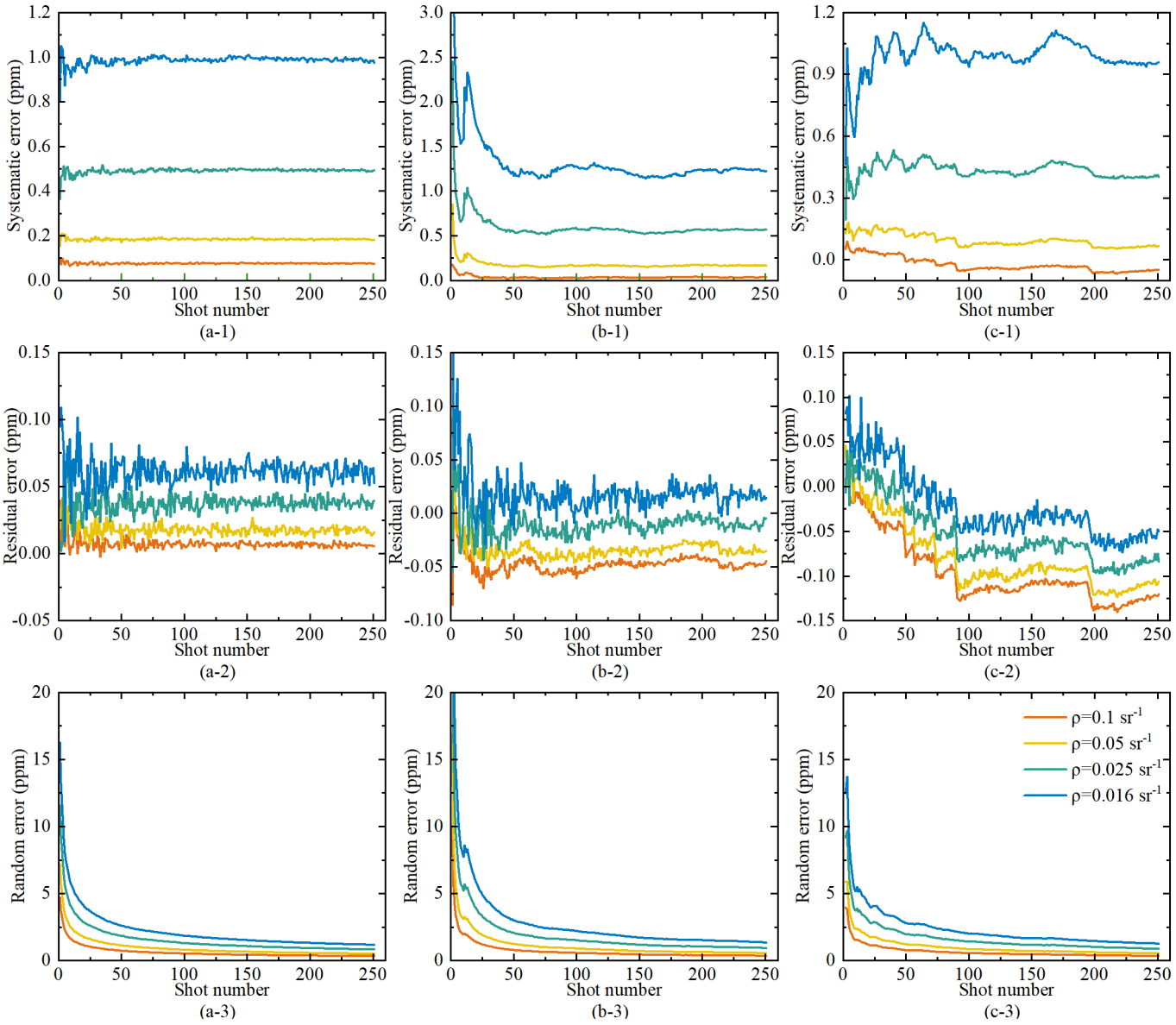


Fig. 6. Distribution of systematic and random errors for the three cities. Different colors represent different surface reflectance. (a-1)–(a-3) Systematic errors, residual systematic errors, and random errors due to random noise in Fuyang, respectively. (b-1)–(b-3) Same as (a-1)–(a-3) except for Huaihua. (c-1)–(c-3) are for Ya'an.

scheme is carried out on the original echo signals, which can directly counteract the random noise of a single shot, resulting in a higher equivalent SNR. However, the systematic errors of the AVS method increase significantly in Ya'an and exceed that of AVX and AVD methods, indicating that the AVS method is easily affected by topographic variations. On the contrary, the systematic errors of AVX and AVD methods are insensitive to geophysical variations.

The systematic errors for the three averaging schemes are corrected based on estimations of SNR at both online and offline wavelengths. The residual systematic errors on CO_2 measurements after bias correction are demonstrated in Fig. 7(b). The surface reflectance is the same as in Fig. 7(a). For AVD and AVX methods, a significant reduction of systematic error is found in every scene, approximately 90%. Since the SNR is estimated from a noisy calibration signal, which is not completely accurate, the systematic errors cannot be fully eliminated. However, only a small part of the systematic errors of the AVS method are corrected, due to the high equivalent SNR. The residual bias of the AVS method mainly comes from geophysical variation, which cannot be removed by the above bias correction.

A comparison of different averaging methods for a surface with a mean reflectance of 0.05 sr^{-1} is discussed above, which roughly corresponds to mixed sea and vegetation cover. Furthermore, surface reflectance is also an important factor for the CO_2 retrievals. High reflectance results in low random errors and higher SNR of observations. To assess the robustness of several averaging methods, the systematic errors for CO_2 detection at different surface reflectances were also estimated. Fig. 8 shows the systematic errors of three averaging methods for four different surface reflectances: 0.1,

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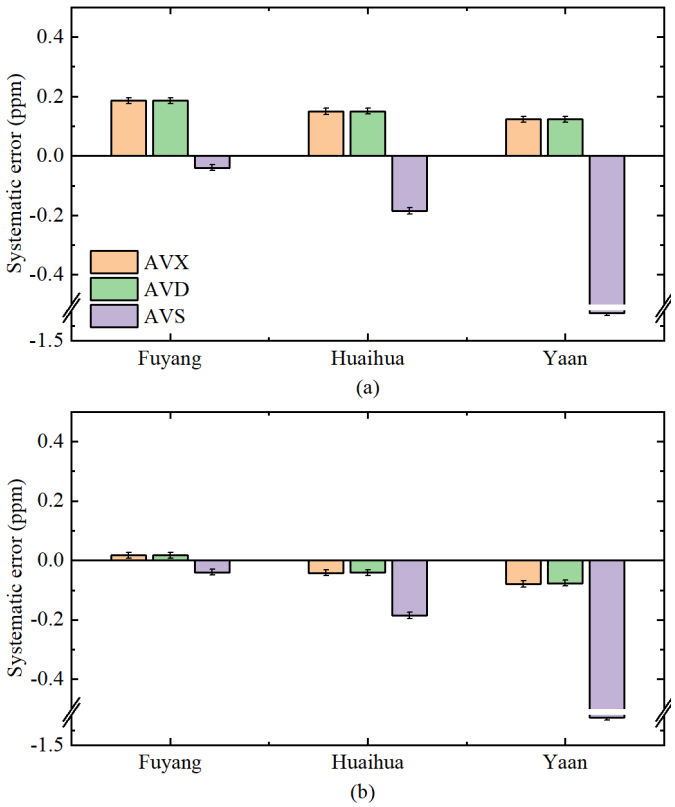


Fig. 7. Systematic errors (a) before and (b) after bias corrections for the three averaging methods at three sites.

0.05, 0.025, and 0.016 sr^{-1}. Compared to a surface reflectance of 0.1 sr^{-1}, the systematic errors for AVX and AVD methods increase for all three cities when the reflectance is 0.05 sr^{-1}. However, the AVX and AVD method still outperforms the AVS methods in Ya'an. The systematic errors at a surface reflectance of 0.025 sr^{-1} show a similar trend to that of 0.05 sr^{-1}. The systematic errors continue to rise for AVD for AVX schemes, while the system errors of the AVS scheme remain stable. This suggests that AVX and AVD schemes are sensitive to reflectance variations, whereas the AVS schemes are less affected by reflectance variations. When the surface reflectance is degraded to 0.016 sr^{-1}, the SNR at online and offline wavelength is 13.7 and 34.6 in Ya'an, respectively. The systematic bias for AVD and AVX methods is increased to 1.18 ppm due to the lower SNR of signals. The systematic errors of the three average methods gradually approach the same value in Ya'an.

The residual systematic errors for the three averaging methods are presented in Fig. 9. Generally, the residual errors of the AVX and AVD methods are comparable. The residual errors of the AVX and AVD methods are basically within 0.1 ppm after bias correction even on the surface with ice and snow cover, indicating the validity of the methodology of bias corrections. The residual bias of the AVS methods is less than the AVX and AVD methods for all reflectance in Fuyang. Consequently, the AVS scheme performed well in Fuyang for four different surface reflectances, with lower systematic errors of 0.02 ppm. However, the AVS method shows higher residual errors in Huaihua and Ya'an than the AVS method due to the high topographic variation. Therefore, the AVX and AVD methods are more suitable for the surface with moderate and severe geophysical variations. The AVS method is more suitable for the surface with slight geophysical variations.

D. Comparison of Averaging Schemes for Observations

In this section, a comparison of three averaging schemes was conducted on the ACDL observations from April to June 2023 to validate against co-located TCCON observations at the Xianghe station. The XCO_2 retrieved from ACDL measurements is the average result of 150 pulses along the satellite flight track, corresponding to a horizontal resolution of about 50 km. To ensure the average number of pulses, the colocated criteria were set as being within a latitude of 1°, a longitude of 1°, and a time window of around 1 h. In addition, the observations at the Xianghe station represent the background state of CO_2 concentration. Consequently, the measurements of ACDL over the city are avoided during the validation.

The XCO_2 retrievals of IPDA measurements for a single shot at the Xianghe station on May 13 are shown in Fig. 10. Fig. 10(a) demonstrates the flight path of the DQ-1 satellite near the Xianghe station. The colors represent the column concentration of CO_2 using the AVS method. Only pixels under the clear sky were included. The XCO_2 ranges from approximately 418 to 430 ppm. It is obvious that the XCO_2 observed by IPDA is higher in residential areas due to large carbon emissions from human activities. The lower XCO_2 is attributed to the photosynthesis of plants. The results demonstrate the capability of ACDL measurements to resolve CO_2 horizontal gradients in the atmosphere. The TCCON XCO_2 at the Xianghe station is shown in Fig. 10(b). The colocated TCCON and ACDL time window is from 5:15:50 to 5:16:10 A.M., corresponding to an XCO_2 of 423.31 ppm at the Xianghe station. Fig. 10(c) presents the XCO_2 retrievals of IPDA measurements for a single shot near Xianghe station. The color represents the point density. The scatters between 5:16:10 and 5:16:40 are relatively sparse because observations from optically thick clouds were removed. The standard deviation of XCO_2 is relatively large. Most of the XCO_2 was in the range of 410–430 ppm. For the three averaging methods, the mean XCO_2 averaged over 150 pulse pairs are 423.92, 423.91, and 423.12 ppm, respectively. The XCO_2 measurements are closer to the observations at the Xianghe station. The bias correction was also performed as described in Section II. After bias correction, the XCO_2 column became 423.72, 423.71, and 423.11 ppm, respectively. The AVS method yielded the lowest bias of 0.2 ppm.

Fig. 11 presents the comparison of XCO_2 retrievals from ACDL observations with TCCON measurements from April to June 2023 at the Xianghe station. The stars represent the colocated TCCON measurements. The dashed bars denote the original mean XCO_2, and the solid bars are the XCO_2 after bias correction. The XCO_2 observed by the TCCON measurements in April and May was about 3 ppm higher than

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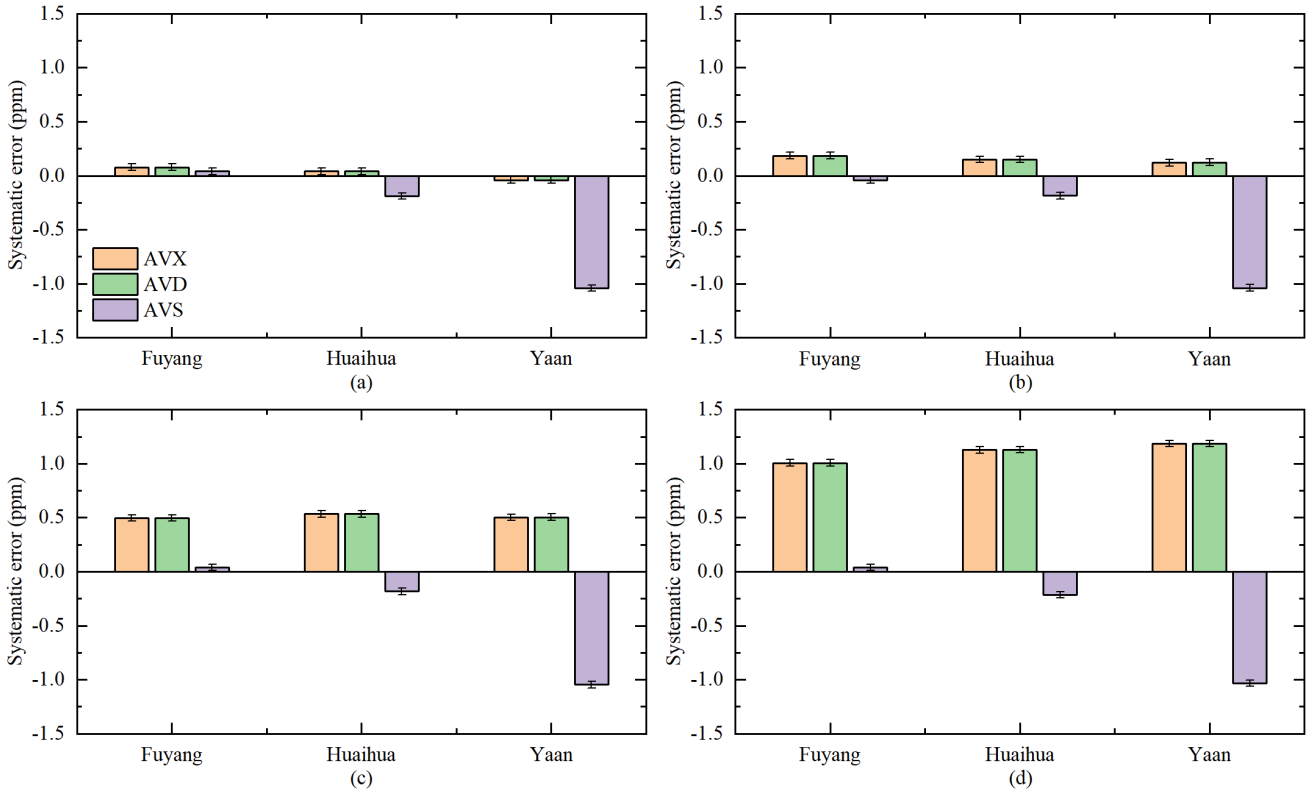


Fig. 8. Systematic errors before bias correction for four surface reflectances. (a)–(d) Systematic errors for 0.1, 0.05, 0.025, and 0.016 sr^{-1}, respectively.

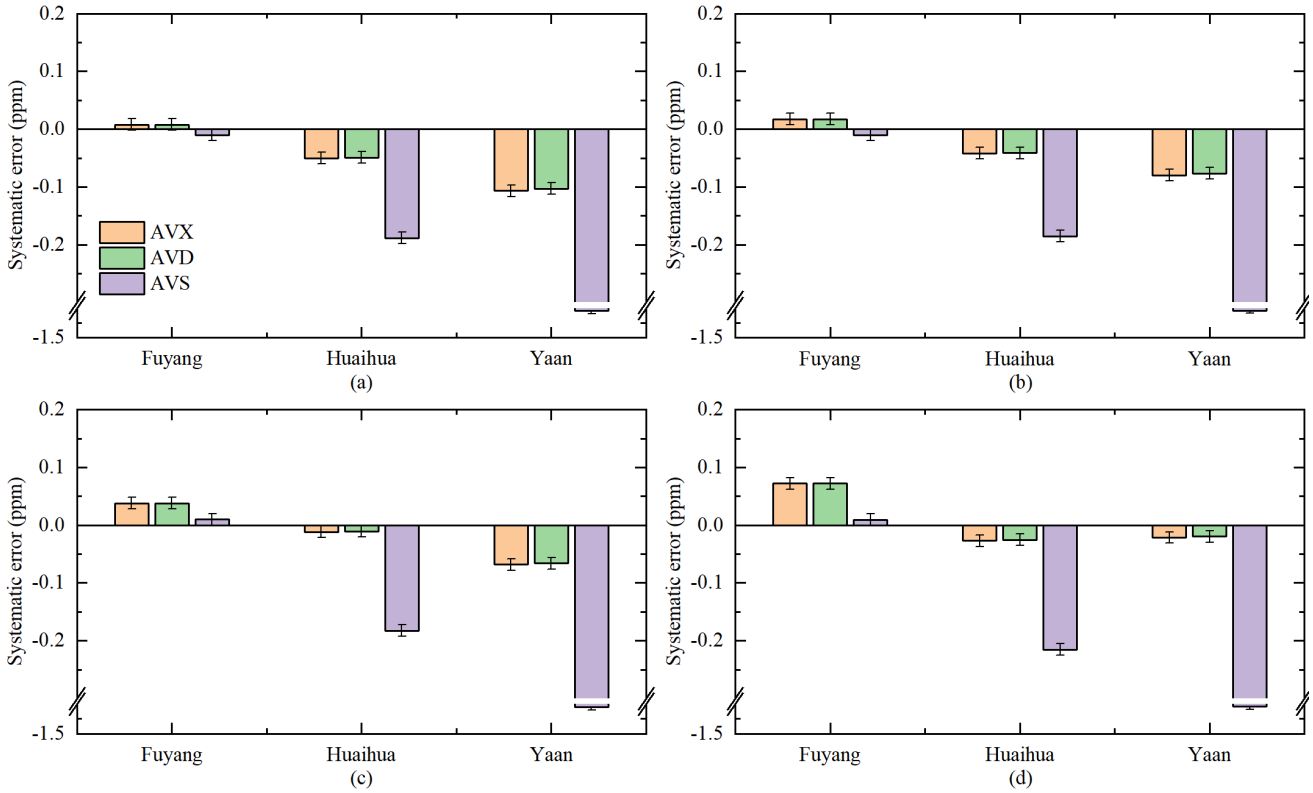


Fig. 9. Residual systematic errors after bias correction for four surface reflectances. (a)–(d) Systematic error for 0.1, 0.05, 0.025, and 0.016 sr^{-1}, respectively.

that of June due to the large anthropogenic sources of CO_2 and the low rate of photosynthesis. The accuracies of the AVD and AVX methods exhibit a significant improvement after averaging bias correction, indicating that the bias correction methods are reliable. For the AVS method, there is little difference between the raw XCO_2 and the XCO_2 after bias

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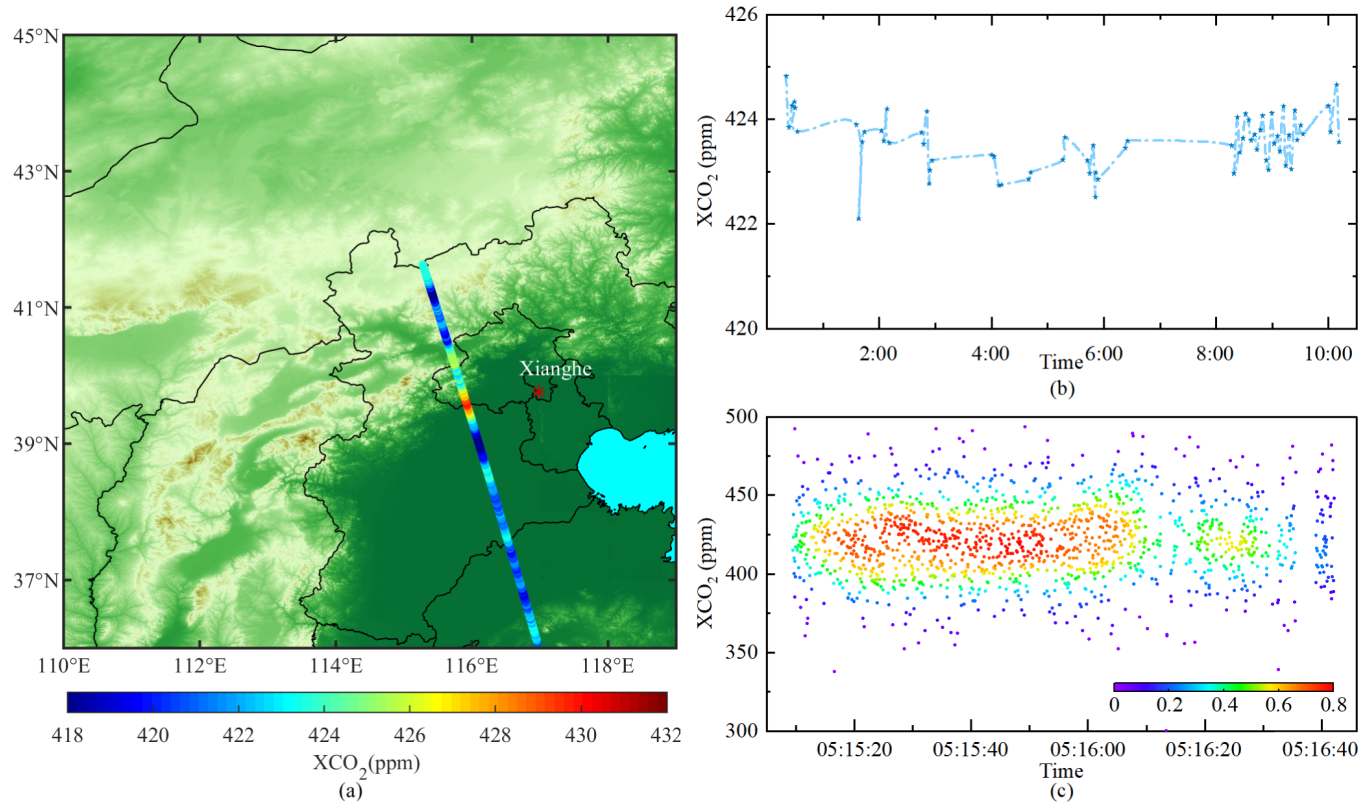


Fig. 10. XCO_2 measurements at the Xianghe station on May 13, 2023. The red star denotes the location of the Xianghe station. (a) Flight path of the DQ-1 satellite. The background is filled with DEM. (b) and (c) XCO_2 observed by Bruker 125HR and ACDL instruments at the Xianghe station, respectively.

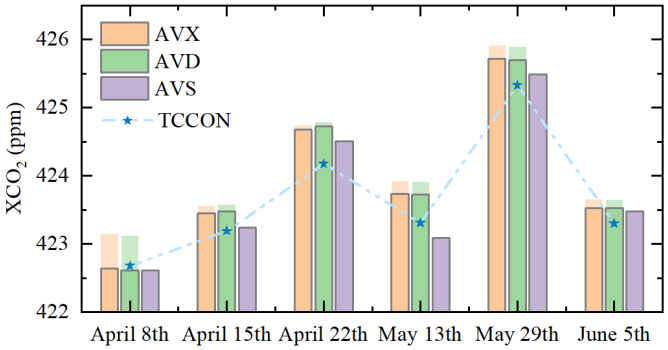


Fig. 11. Comparison of XCO_2 retrievals from ACDL measurements with TCCON measurements from April to June 2023 at the Xianghe station.

correction, which is consistent with the simulation results. The XCO_2 retrievals for all three averaging schemes have a good agreement with the TCCON measurements. The biases between IPDA XCO_2 retrievals and TCCON observation at Xianghe station are 0.52, 0.52, and 0.34 ppm for AVX, AVD, and AVS methods, respectively. The CO_2 retrievals for ACDL measurements with the AVS method exhibit the lowest bias.

The above experiment indicates that the AVS method is suitable for CO_2 retrieval at the Xianghe station, where the surrounding topography is flat. The XCO_2 retrieved from ACDL measurements is credible with an accuracy of less than 0.5 ppm, which offers valuable insights into quantifying the magnitude and distribution of terrestrial and oceanic sources and sinks.

V. Conclusion and Discussion

The XCO_2 retrievals from the IPDA LiDAR are affected by random noise, which is usually reduced by averaging over several pulse pairs. However, the nonlinearity of the IPDA LiDAR measurement equation will result in a systematic bias in the averaging process. Therefore, a reasonable averaging scheme for ACDL measurements is needed to balance the random and systematic errors. In this article, the performance of three averaging schemes (AVX, AVD, and AVS) was compared to provide an optimum averaging method for ACDL measurements. Experiments are carried out on both simulations and ACDL observations. The random and systematic errors for a single shot due to the random noise of the ACDL instrument are estimated based on simulations. In addition, the influence of reflectance and altitude on the systematic and random errors is analyzed. It is found that the random and systematic errors of the CO_2 retrievals are inversely proportional to the surface reflectance and the altitude of the echo signals. The single-shot bias due to random noise is in the range of 0–0.8 ppm and can be reduced to 0.18 ppm after bias correction. The precision for a single shot ranges from 1.5 to 19 ppm, which does not meet the requirements of CO_2 measurement, indicating the necessity of averaging during the retrievals. The CO_2 retrievals for averaging pulse pairs are affected by the averaging window size and the averaging method. The optimal size of the averaging window is determined based on the sensitivity analysis of CO_2 retrievals at different surfaces. An averaging of measurements over 150 pulse pairs proves to

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be more appropriate, corresponding to an averaging window size of 50 km. The accuracy for averaging pulse pairs after bias correction is better than 0.05 ppm with an average window size of 50 km. The precision is significantly improved to within 0.36% compared to single-shot measurements. Then, the performance of three averaging schemes was compared at three sites with different topographical variabilities by evaluating the accuracy of CO_2 retrievals. The results show that the AVX and AVD schemes show similar performance in all three scenarios. Both perform well in Huaihua and Ya'an with a residual bias of less than 0.1 ppm. The AVS scheme exhibits better performance in Fuyang even in the lower reflectance of 0.016 sr^{-1}. The residual bias is within 0.02 ppm. It is found that the AVX and the AVD schemes are less sensitive to altitude variations. Therefore, the AVX and the AVD schemes are more suitable for surfaces with moderate and severe topographic variations. In contrast, the AVS method is easily affected by altitude variations but more robust to surface reflectance due to the high equivalent SNR. Thus, the AVS method is more applicable to surfaces with slight topographic variations, especially for low surface reflectance. This is slightly different from the averaging scheme of Methane Remote Sensing LiDAR Mission (MERLIN), a future spaceborne methane IPDA LiDAR. For the MERLIN mission, the AVS method performs well on surfaces with both slight and moderate topographic variations. This may be due to the different concentrations of CH_4 and CO_2. In conclusion, for the ACDL instrument, the AVX and AVD methods are recommended when conducting observations over hills and mountains. Conversely, the AVS method is preferable for observations over oceans, plains, and terraces. Furthermore, a comparison of three averaging schemes is also carried out on actual ACDL observations. The CO_2 retrievals from ACDL observations from April to June 2023 were analyzed. The XCO_2 averaged over 150 pulse pairs have a good agreement with the TCOON measurements. The lowest biases between CO_2 retrievals and TCCON observations are obtained from the AVS scheme of less than 0.5 ppm, which is consistent with the simulation results.

The averaging scheme is an essential step in CO_2 retrievals for ACDL observations. The performance of the averaging scheme varies across different surfaces. This article provides a more comprehensive averaging scheme for ACDL observations over several surfaces compared to studies based on airborne measurements conducted at a specific site [35]. In addition, the systematic errors in the averaging process due to the nonlinearity of the IPDA LiDAR measurement equation are also taken into account. This study helps to provide a reasonable averaging scheme for ACDL observations. It can serve well the downstream applications and retrievals for ACDL measurements. Presently, the validation of the CO_2 retrievals has only been carried out at the Xianghe station due to the limited ground-based observations. In the future, the comparison of three averaging schemes will be conducted at more ground-based sites. The experiments are carried out on the clear sky simulations. However, there may be partial clouds in the averaging window. Therefore, a reasonable averaging scheme is needed to be further investigated in the case of a partially cloudy window.

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Xifeng Cao was born in Shandong, China, in 1996. She received the B.S. degree in aerospace engineering from the Shandong University of Science and Technology, Qingdao, China, in 2018, and the M.S. degree in electronics and communication engineering from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2021. She is currently pursuing the Ph.D. degree in atmospheric science with Fudan University, Shanghai, China.

Her research interests include light detection and ranging (LiDAR) data processing, retrieval algorithm of greenhouse gas, and CO_2 emission.



Xingying Zhang was born in Fujian, China, in 1978. He received the B.S. degree in polymer materials and engineering from Beihang University, Beijing, China, in 2001, and the Ph.D. degree in physical chemistry from Beijing Normal University, Beijing, in 2006.

From 2018 to 2020, he was the Director of the Business Science and Technology Division, National Satellite Meteorological Center (National Center for Space Weather), Innovation Center for FengYun Meteorological Satellite (FYSIC), China Meteorological Administration (CMA), Beijing. From 2020 to 2021, he was the Director of the Satellite Meteorology Institute, National Satellite Meteorological Center (National Center for Space Weather). From 2021 to 2022, he was a Deputy Director of the National Satellite Meteorological Center (National Center for Space Weather). Since 2022, he has been the Deputy Director of the Department of Science, Technology and Climate Change, CMA. He is the author of five academic monographs, nine national and industry standards, and more than 120 articles. He hosted more than 40 scientific research projects such as the National Natural Science Foundation, the 863 Program, the 973 Program, the Science and Technology Support Plan, and industry special projects. His research interests include atmospheric chemical composition monitoring and its impact on the atmospheric environment and climate change.



Lu Zhang was born in Shaanxi, China, in 1988. He received the Ph.D. degree in atmospheric remote sensing from the Nanjing University of Information Science and Technology, Nanjing, China, in 2019.

He is currently a Senior Engineer with the National Satellite Meteorological Center (National Center for Space Weather), Innovation Center for FengYun Meteorological Satellite (FYSIC), China Meteorological Administration (CMA), Beijing, China. His area of interest is remote sensing of greenhouse gases.



Minqiang Zhou received the B.S. degree from the Nanjing University of Information Science and Technology, Nanjing, China, in 2012, and the Ph.D. degree from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, in 2017.

From 2015 to 2017, he was a Research Assistant with the Royal Belgian Institute for Space Aeronomy (BIRA-IASB), Uccle, Belgium. From 2017 to 2019, he was a Scientist SW11 with BIRA-IASB. Since 2019, he has been an Associate Researcher with the Institute of Atmospheric Physics, Chinese Academy of Sciences. He is the author of more than 25 articles. His research interests include atmospheric remote sensing and atmospheric environment detection.



Yuhan Jiang was born in Sichuan, China, in 1999. She received the B.S. degree from the School of Atmospheric Science, Chengdu University of Information Science and Technology, Chengdu, China, in 2022. She is currently pursuing the M.S. degree with the Chinese Academy of Meteorological Sciences, Beijing, China.

Her research interests include remote sensing of atmospheric greenhouse gases and their climate effects.



Jiqiao Liu received the B.S. degree from Wuhan University, Wuhan, China, in 2001, and the Ph.D. degree from the Shanghai Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Shanghai, China, in 2006.

From 2006 to 2008, he was a Research Assistant with the Shanghai Institute of Optics and Precision Machinery, Chinese Academy of Sciences. From 2008 to 2014, he was an Associate Researcher with the Shanghai Institute of Optics and Precision Machinery, Chinese Academy of Sciences. In 2013, he was a Visiting Scholar with the Leibniz Institute of Atmospheric Physics, Kühlungsborn, Germany. Since 2014, he has been a Researcher with the Shanghai Institute of Optics and Precision Machinery, Chinese Academy of Sciences. He is the author of more than 116 articles and more than 39 patents. His research interests include new laser remote sensing technology for atmospheric remote sensing, atmospheric wind field detection light detection and ranging (LiDAR), atmospheric greenhouse gas detection LiDAR, and atmospheric aerosol and cloud detection LiDAR.

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Tiantao Cheng received the B.S. degree in climatology from the Nanjing University of Information Science and Technology, Nanjing, China, in 1997, the M.S. degree in meteorology and climatology from the Institute of the Cold and Arid Regions Environment and Engineering, Lanzhou, China, in 2000, and the Ph.D. degree in atmospheric physics and atmospheric environment from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, in 2003.

From 2004 to 2006, he was a Lecturer with Fudan University, Shanghai, China. From 2006 to 2007, he was a Visiting Scholar with the Max-Planck-Institute for Meteorology, Hamburg, Germany. From 2006 to 2011, he was an Associate Professor with Fudan University. Since 2011, he has been a Professor with Fudan University. He is the author of more than 23 articles. His research interests include optical properties of aerosols and direct radiative forcing; changes in microscopic physical and chemical properties of aerosol particles and cloud condensation nucleus (CCN) activity, aerosol-cloud-precipitation interactions; the effects of atmospheric pollution on regional haze, fog, clouds, and the influence of precipitation formation and long-term trends; and changes in the thermodynamic structure within the urban atmospheric boundary layer and the interaction between pollutants and meteorological factors.



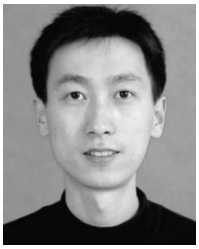
Chuncan Fan was born in Shandong, China, in 1997. He received the B.S. degree in aerospace engineering from the Changchun University of Science and Technology, Changchun, China, in 2019. He is currently pursuing the Ph.D. degree in physics with the Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Beijing, China.

His research interests include atmospheric greenhouse gas detection light detection and ranging (LiDAR), LiDAR data processing, and retrieval algorithm of greenhouse gas.



Cheng Chen was born in Zhejiang, China, in 1998. She received the B.S. degree in optoelectronic information science and engineering from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2020. She is currently pursuing the Ph.D. degree in optical engineering with the Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences, Beijing, China.

Her research interests include light detection and ranging (LiDAR) for atmospheric greenhouse gas detection and retrieval.



Lin Chen received the Ph.D. degree in atmospheric physics and environment from the Chinese Academy of Sciences, Beijing, China, in 2009.

He is currently an Associate Researcher with the National Satellite Meteorological Center (National Center for Space Weather), Innovation Center for FengYun Meteorological Satellite (FYSIC), China Meteorological Administration (CMA), Beijing. He is the author of more than 81 articles. His research interests include instrument development and design on the early morning orbit satellite, calibration of optical meteorology satellites, and retrieval of aerosols' properties.

Dr. Chen is a member of the Global Space-Based Inter-Calibration System/World Meteorological Organization (GSICS/WMO).

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